

# VajraGrid

AI-Driven Cyber Defense for Smart Power Grids

**AMD**  
Slingshot

**HUMAN** *IMAGINATION*  
**BUILT WITH AI**

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## Team Details

**Team:** Innovibe

**Lead:** D Shantan Dheer

**Members:** MJ Emmanuel, Jakkula Tarun

**Problem:** Smart Grid Cybersecurity & AI Defense

## Brief About the Idea

**575+ GW**

India's Grid — 3rd Largest

**₹100+ Cr**

Cost Per Hour of Downtime

**ZERO**

AI Cyber Defense for Indian Grid

### The Problem

- 2020 Mumbai blackout: Chinese malware in SCADA
  - 2023 ransomware: zero automated response
  - 20 Cr smart meters = new attack surfaces
- No AI cyber defense exists for Indian grids

### VajraGrid Solution

- 4-layer AI detection (Rules → Physics → Stats → ML)
  - VajraShield™ autonomous FLISR healing
    - <3s detection, 16s full grid restoration
- Edge-first on AMD Ryzen AI NPU

VajraShield™ detects attacks in <3 seconds and restores grid in 16 seconds — no human intervention

## Opportunities & Differentiation

### How Is It Different?

Existing tools (OSIsoft PI, ABB FOX615) are passive monitors.

VajraGrid is the FIRST to combine:

- Multi-layer AI detection
- Autonomous grid healing
- Edge NPU deployment
- India-specific grid topology

### How Does It Solve?

Manual response: 30-90 minutes.  
VajraGrid: 3s detect + 16s heal.

- Zero false positive design
- Works on airgapped substations
- Scales to 250K+ substations
- ₹2-5L per edge node

### USP

- 4-Layer Fusion Detection
- Physics-aware (Kirchhoff validation)
- Self-healing FLISR algorithm
- AMD Ryzen AI NPU optimized
- Real working prototype
- ₹100+ Cr/incident savings

## Key Features

### 4-Layer Detection

Rules (9) → Physics → Stats → ML/ONNX. Each <500ms.

### AlertClassifier Fusion

Multi-signal confidence → triggers VajraShield at 0.90+

### VajraShield™ FLISR

Fault locate → isolate → reroute → restore in 16s

### 5 Attack Types

FDI, Command Spoof, MaDIoT, Sensor Tamper, Smart Meter

### SOC Dashboard

Live telemetry, topology map, threat timeline, alert feed

### Topology Agnostic

5-bus demo scales to 10K+ node state grids

### Edge Deployment

ONNX on AMD Ryzen AI NPU — 50 TOPS, 15W cTDP

### Mobile Responsive

Full SOC on mobile: drawer nav, compact cards

## Detection & Healing Pipeline

DATA FLOW: SCADA Telemetry (1s) → API → DataGenerator → 4 Detection Engines → AlertClassifier → VajraShield

### Rule Engine

9 rules

<100ms

### Physics Engine

4 Kirchhoff checks

<100ms

### Statistical

Z-score + CUSUM

<200ms

### ML/ONNX

Isolation Forest

<500ms

ALERTCLASSIFIER FUSION: All 4 layers → Multi-signal confidence score  $\geq 0.90$  → VajraShield™ trigger

DETECTING

ISOLATING

REROUTING

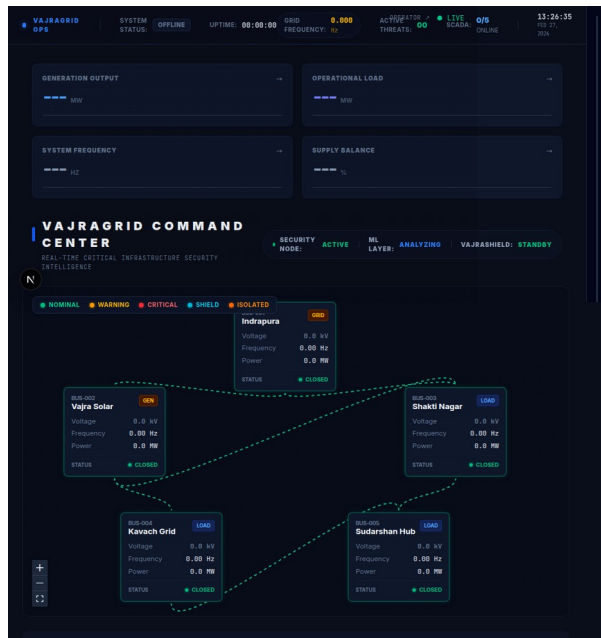
MONITORING

RESTORING

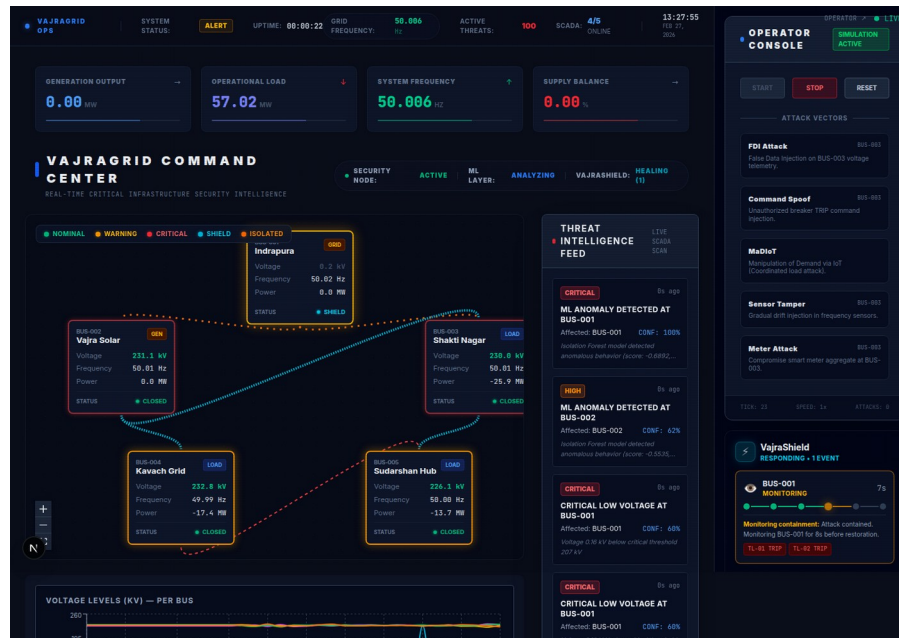
RESTORED

TIMING: Detection <3s | Full Heal Cycle: 16s | Manual baseline: 30-90 min

## Live Dashboard



**Dashboard at Startup**  
Topology map, metric cards, dark SOC theme



**Live Simulation + Healing**  
Real-time telemetry, alerts, VajraShield FLISR

# System Architecture

## PRESENTATION

Next.js 16 + React 19 | Tailwind v4 | Recharts + Topology Canvas

## API LAYER

REST: /api/simulation/{start,stop,attack} | SSE: /api/simulation/stream

## SIMULATION ENGINE

DataGenerator → Telemetry (V, f, P, Q,  $\theta$ , AMI) | 5 Attack Injectors

## 4-LAYER DETECTION

Rules (9) → Physics (4) → Statistical (Z+CUSUM) → ML/ONNX (IsolationForest)

## VAJRASHIELD™

FLISR: Fault Location → Breaker Isolation → Power Rerouting → Restoration

## AMD HARDWARE

Edge: Ryzen AI NPU (XDNA 50 TOPS) + VitisAI | Dev: EPYC AVX2+FMA3

## Technology Stack

| Layer           | Technology                                                  |
|-----------------|-------------------------------------------------------------|
| Frontend        | Next.js 16, React 19, Tailwind v4, Recharts, D3.js          |
| Backend         | Next.js API Routes, Server-Sent Events (SSE)                |
| AI/ML           | ONNX Runtime (Isolation Forest), 4-Layer Detection Pipeline |
| Simulation      | Custom DataGenerator, 5 Attack Models, Physics Engine       |
| Self-Healing    | VajraShield FLISR Algorithm (6 phases)                      |
| Hardware Target | AMD Ryzen AI NPU (XDNA), Vitis AI EP, Quark INT8            |
| Dev Platform    | AMD EPYC (AVX2+FMA3), GitHub Codespaces                     |
| Testing         | Vitest (37/37), ESLint, TypeScript Strict                   |



## AMD Products & Solutions

### AMD Ryzen™ AI NPU — XDNA

- Edge inference at substations
- 50 TOPS sustained, 15W cTDP
- Runs anomaly\_detector.onnx locally
- <5ms inference vs 200ms cloud

### ONNX + Vitis AI EP

- Model already in ONNX format
- VitisAI Execution Provider for NPU
- Quark INT8 quantization (<1% loss)
- Zero framework changes needed

### AMD EPYC — Dev Platform

- Current prototype on EPYC/AVX2
- 31,500 inferences/sec achieved
- OpenBLAS DYNAMIC\_ARCH support
- Cloud dev → edge deployment path

### Zen 5 CPU — Heterogeneous

- Rules/Physics/Stats on CPU cores
  - ML offloaded to NPU
  - Parallel processing pipeline
- ROCm GPU path for training

Edge node: ~₹2-5L/substation | Up to 5-8x inference speedup (NPU vs CPU) | 15W cTDP

## Commercial Impact & ROI

**\$8.3B**

Grid Market 2025

**\$22.7B**

2034 @ 11.7% CAGR

**11.7%**

Smart Grid Cyber CAGR

**<1 Event**

Payback Period

| Revenue Stream      | Pricing                     |
|---------------------|-----------------------------|
| SaaS per-substation | ₹5,000/month/node           |
| Enterprise license  | ₹50L-1Cr/year per DISCOM    |
| Edge hardware       | ₹2-5L per AMD Ryzen AI node |

### ROI Case

- Avg blackout: ₹100+ Cr/incident
- VajraGrid: <3s detect, 16s heal
  - Manual: 30-90 min response
  - Risk reduction: ~99%
- Payback: <1 prevented event

Customers: State DISCOMs (BESCOM, MSEDCL, TPDDL) | PGCIL | Smart Meter Vendors | Industrial

## Prototype Assets

### GitHub Repository

Primary: [github.com/vajragrid](https://github.com/vajragrid)  
Mirror: [github.com/D-Shantanu-Dheer/vajragridr](https://github.com/D-Shantanu-Dheer/vajragridr)

Tech: Next.js 16 + ONNX Runtime  
License: MIT Open Source

### Demo Walkthrough

1. Open VajraGrid dashboard
2. Click "Start Simulation"
3. Observe live telemetry
4. Trigger attack → watch detection
5. VajraShield auto-heals grid

37/37

Tests Pass

0

Warnings

95.8%

ML Rate

0.2%

False +ve

## Scalability & Roadman

| Component         | 5-Bus (Demo)  | 100-Bus       | 10K+ (State)   |
|-------------------|---------------|---------------|----------------|
| Buses/Nodes       | 5             | 100           | 10,000+        |
| Detection Latency | <3s           | <5s           | <10s           |
| Heal Cycle        | 16s           | 30s           | 60s            |
| Hardware          | 1 Ryzen AI    | 10 edge nodes | 250K+ nodes    |
| Deployment        | Single campus | City DISCOM   | State/National |

### Near-Term

- OPC-UA SCADA adapter
- Real telemetry integration
- CERC compliance module

### Mid-Term

- GIS Mapbox dashboard
- Multi-DISCOM federation
- AMI head-end protection

### Long-Term

- Commercial SaaS platform
- International grid standards
- ROCm GPU training pipeline

## Attack Detection — How It Works

### Layer 1: Rule Engine

9 predefined rules check voltage ( $\pm 5\%$ ), frequency ( $\pm 0.5\text{Hz}$ ), ROCOF, breaker status, power factor. Deterministic,  $<100\text{ms}$ . Catches command spoofing and sensor tampering instantly.

### Layer 2: Physics Engine

4 physics-based validations: power balance ( $\Delta P < 5\%$ ), Kirchhoff current law, reactive power coupling, voltage-angle consistency. Catches FDI and MaDIoT attacks.

### Layer 3: Statistical Engine

Z-score outlier detection + CUSUM change detection. Cross-bus correlation analysis. Catches slow drift attacks and ramp manipulations.

### Layer 4: ML/ONNX

Isolation Forest trained on normal+attack patterns. ONNX Runtime inference. 95.8% accuracy, 0.2% false positive. Catches novel zero-day attacks.

## VajraShield™ — Autonomous Grid Healing

### DETECTING (0-3s)

AlertClassifier fuses all 4 layers. When combined confidence  $\geq 0.90$ , VajraShield activates. Identifies affected bus segment.

### ISOLATING (3-5s)

Opens breakers on compromised bus. Prevents cascading failures. Logs isolation event to SOC dashboard.

### REROUTING (5-10s)

Calculates alternate power paths. Transfers load to healthy buses. Maintains grid stability during transition.

### MONITORING (10-13s)

Validates grid stability post-reroute. Checks voltage, frequency, power balance on all buses. Confirms no secondary faults.

### RESTORING (13-16s)

Gradually brings isolated segment back online. Re-closes breakers. Confirms full grid restoration. Total: 16 seconds.



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**Thank you!**

